DFT's missing energy: same density, different electron-pair probabilities.

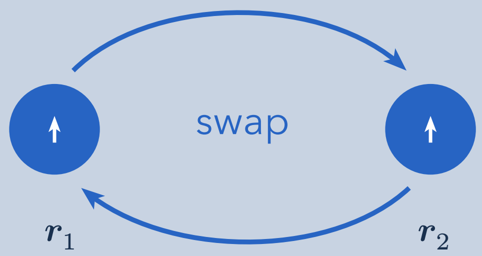
A **functional** maps the whole density to an energy. Kohn-Sham uses **noninteracting electrons at the same density**.

$$E[n] = \underset{\text{KS kinetic energy}}{T_s} + \underset{\text{nuclear attraction}}{V_{\text{ext}}} + \underset{\text{smooth-charge repulsion}}{E_H} + \underset{\text{the missing correction}}{E_{\text{xc}}}$$

$$E_{\text{xc}} = \underbrace{T - T_s}_{\text{kinetic correction}} + \underbrace{V_{ee} - E_H}_{\text{interaction correction}} = E_x + E_c$$

EXCHANGE

Antisymmetry: swapping fermions changes the sign.



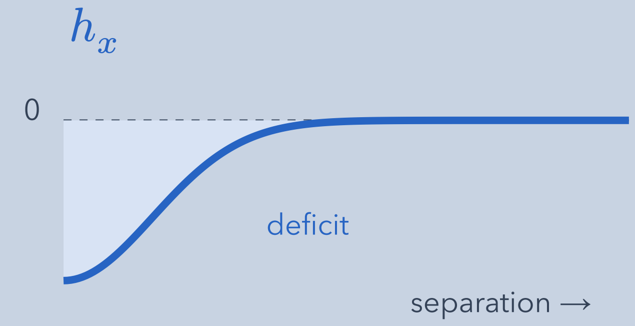
$$\Psi \longrightarrow -\Psi$$

same position $\longrightarrow \Psi = 0$

Pauli exclusion • same spin only • present in one determinant

Opposite-spin pairs have no exchange hole.

same-spin probability



$$\int h_x \mathrm{d}^3\boldsymbol{r}' = -1$$

One missing electron. No extra force.

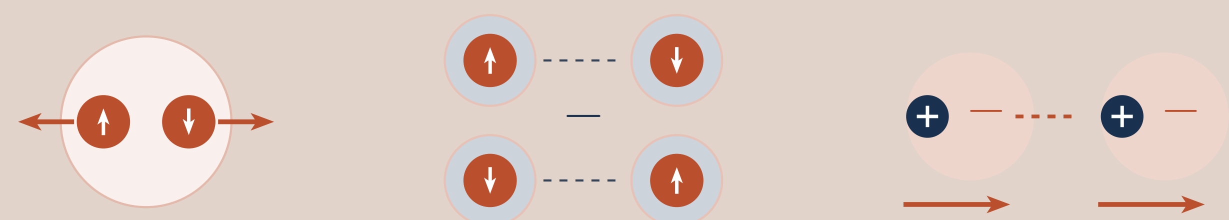
The hole persists without Coulomb repulsion.

$$E_x = \langle \Phi_s \mid \hat{V}_{ee} \mid \Phi_s \rangle - E_H \leq 0$$

Φ_s: antisymmetric KS orbital state. T, V_{ee}: exact interacting values at the same density.

CORRELATION

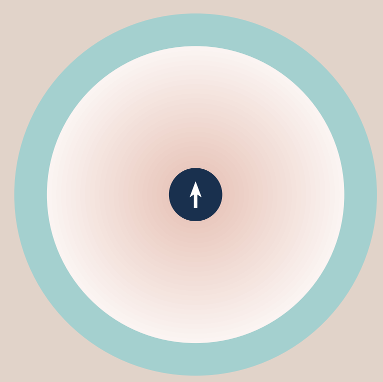
Joint probabilities beyond the KS determinant.



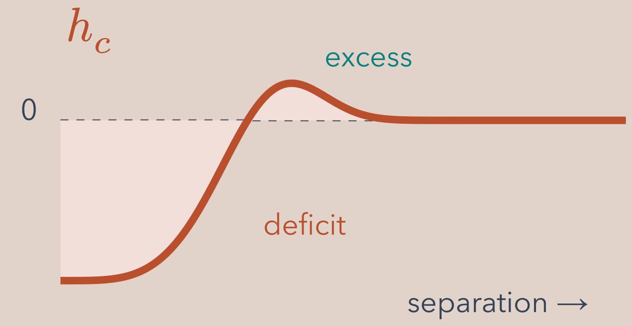
DYNAMIC
short-range avoidance and screening

STATIC
several configurations stretched-bond singlet

DISPERSION
coupled fluctuations even at zero temperature



change in probability



$$\int h_c \mathrm{d}^3\boldsymbol{r}' = 0$$

Same-spin and opposite-spin pairs.

A near deficit is balanced by an excess farther away.

$$E_c = \underbrace{T - T_s}_{T_c \geq 0} + \underbrace{V_{ee} - E_H - E_x}_{U_c} \leq 0$$

Correlation balances reduced repulsion against a kinetic cost.

ONE REFERENCE ELECTRON

Only $N-1$ others remain.

$$n_{\text{cond}} = n + h_x + h_c$$

net hole: $-1 + 0 = -1$

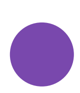
probability deficit, not an empty cavity

CLASSICAL CHARGES

Correlation: **yes**
Pauli exchange: **no**

correlated positions

ONE ELECTRON: NO SELF-REPULSION



$$E_H + E_x = 0$$
$$E_c = 0$$

Approximate XC can violate this cancellation.

XC FEEDS BACK INTO THE DENSITY

$v_{\text{xc}} = \frac{\delta E_{\text{xc}}}{\delta n} \longrightarrow$ orbitals $\longrightarrow n$

Why approximate? Exact XC is formally defined, but no practical general expression is known.

LDA: local electron gas · GGA: density gradients · hybrids: orbital exchange. Ordinary LDA/GGA miss long-range dispersion.

Schematic maps / curves; map center = reference electron. Hole sums are 3D integrals. Fixed nuclei; spin labels suppressed; atomic units; nuclear repulsion added separately.

Burke: The ABC of DFT, §§7, 11–13 · The exchange-correlation hole · Archer: Classical fluids, ch. 2